

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

Register Number	192212182
Name	U Fayaz Ahmad

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I U Fayaz Ahmad (192212182) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well- explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

1. SMART HOME AUTOMATION SYSTEM – INTELLIGENT AGENTS

Aim

To explain the different types of intelligent agents and analyze how they function in a smart home automation system to provide comfort, energy efficiency, and security.

Problem Statement

A smart home automation system is designed to manage **lighting, temperature, and security** based on user behavior and environmental conditions. The system receives real-time inputs such as motion detection, temperature changes, and time of day while learning user preferences over time.

The objective is to analyze how different intelligent agents operate in this system and determine the most suitable agent architecture.

1. Simple Reflex Agent

Working

A Simple Reflex Agent makes decisions based only on the **current input (percept)** using predefined condition–action rules.

Examples

- If motion is detected → Turn ON the lights.
- If no motion is detected → Turn OFF the lights.
- If smoke is detected → Activate the fire alarm.
- If the main door opens unexpectedly → Trigger the security alarm.

Advantages

- Fast response.
- Easy to implement.
- Suitable for simple automation tasks.

Limitations

- No memory of previous events.
 - Cannot learn user preferences.
 - Inefficient in complex situations.
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2. Model-Based Reflex Agent

Working

A Model-Based Reflex Agent maintains an **internal model** of the environment and remembers previous states to make better decisions.

Examples

The agent stores information such as:

- Current room temperature.
- Lights already switched ON.
- Door and window status.
- Previous user activities.
- Occupancy of different rooms.

If the living room is already occupied and the temperature falls below the preferred level, the system automatically switches ON the heater.

Advantages

- Maintains memory of past states.
- Handles partially observable environments.
- Improves decision-making.

Limitations

- More complex than a Simple Reflex Agent.
 - Requires additional memory and processing.
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3. Goal-Based Agent

Working

A Goal-Based Agent selects actions that help achieve specific goals.

Goals in Smart Home

- Maintain a comfortable room temperature.
- Reduce electricity consumption.
- Keep the home secure.
- Provide maximum user comfort.

Example

If the desired room temperature is **24°C**, the system decides whether to switch ON the air conditioner or heater to achieve that goal.

Advantages

- Plans actions intelligently.
- Flexible and adaptable.
- Suitable for complex environments.

Limitations

- Requires planning before taking action.
 - Slower than reflex agents.
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4. Utility-Based Agent

Working

A Utility-Based Agent chooses the action that provides the **maximum overall benefit (utility)** by considering multiple factors.

Factors Considered

- User comfort
- Energy consumption
- Electricity cost
- Security
- Weather conditions
- Time of day

Example

During peak electricity hours, the system increases the AC temperature slightly to reduce power consumption while maintaining user comfort.

Advantages

- Optimizes multiple objectives.
- Produces the best overall decision.
- Highly suitable for smart home systems.

Limitations

- Requires utility calculations.
 - Higher computational complexity.
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Comparison of Intelligent Agents

Agent Type	Memory	Goal-Oriented	Learns Preferences	Suitable for Smart Home
Simple Reflex Agent	No	No	No	Basic Automation
Model-Based Agent	Yes	No	Partial	Good
Goal-Based Agent	Yes	Yes	Partial	Better
Utility-Based Agent	Yes	Yes	Yes	Best

Hybrid Intelligent Agent

The most effective approach is a **Hybrid Intelligent Agent**, which combines the features of:

- **Simple Reflex Agent** for immediate responses.
 - **Model-Based Agent** for maintaining environmental information.
 - **Goal-Based Agent** for achieving comfort and security goals.
 - **Utility-Based Agent** for selecting the best action based on energy efficiency, user comfort, and security.
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Justification

A Hybrid Intelligent Agent is the most suitable because it:

- Responds quickly to real-time events.
 - Learns user preferences over time.
 - Optimizes energy consumption.
 - Improves home security.
 - Maintains comfortable environmental conditions.
 - Makes intelligent decisions using multiple criteria.
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Conclusion

A smart home automation system requires intelligent decision-making to balance comfort, energy efficiency, and security. While each intelligent agent has its own strengths, combining them in a Hybrid Intelligent Agent provides the most effective and reliable solution.

Result

Thus, a **Hybrid Intelligent Agent** is the most suitable architecture for a smart home automation system because it integrates the advantages of **Simple Reflex, Model-Based, Goal-Based, and Utility-Based Agents**. It ensures fast responses, adapts to user preferences, minimizes energy consumption, enhances security, and provides maximum comfort through intelligent and efficient decision-making.

2. ROBOT IN AN UNKNOWN MAZE – UNINFORMED SEARCH STRATEGIES

Aim

To explain how **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)** help a robot find the exit in an unknown maze, compare their advantages and disadvantages, and identify the most suitable intelligent agent and search strategy.

Problem Statement

A robot is placed in an **unknown maze** without any prior knowledge of its layout. The robot must explore the maze, locate the exit, and reach it safely. Since no map is available, the robot uses **uninformed search strategies** to explore the environment and make decisions.

Uniform Cost Search (UCS)

Working

Uniform Cost Search expands the node with the **lowest cumulative path cost** first. The robot calculates the total cost from the start node to every reachable node and always chooses the path with the minimum cost.

Example

Suppose the robot has three possible paths:

Path	Cost
$S \rightarrow A \rightarrow G$	8
$S \rightarrow B \rightarrow C \rightarrow G$	5
$S \rightarrow D \rightarrow G$	10

Since the path $S \rightarrow B \rightarrow C \rightarrow G$ has the lowest cost (**5**), UCS selects this path.

Advantages of UCS

- Guarantees the optimal (least-cost) solution.
 - Complete algorithm (finds a solution if one exists).
 - Suitable when movement costs are different.
 - Avoids expensive paths.
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Disadvantages of UCS

- Requires a large amount of memory.
 - Slower for large search spaces.
 - Expands many unnecessary nodes before reaching the goal.
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Iterative Deepening Search (IDS)

Working

Iterative Deepening Search repeatedly performs **Depth-First Search (DFS)** with increasing depth limits until the goal is found.

The search proceeds as:

- Depth = 0
 - Depth = 1
 - Depth = 2
 - Continue increasing the depth until the exit is reached.
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Example

Search Order:

Depth 0:

S

Depth 1:

$S \rightarrow A$

$S \rightarrow B$

Depth 2:

$S \rightarrow A \rightarrow C$

$S \rightarrow B \rightarrow D$

Depth 3:

Continue until Goal (G) is found.

Advantages of IDS

- Uses very little memory.
- Complete algorithm.
- Suitable when the solution depth is unknown.
- Combines the advantages of DFS and BFS.

Disadvantages of IDS

- Repeats node expansions at every depth level.
- Higher execution time because of repeated searches.
- Not suitable when path costs are different.

Time and Space Complexity

Algorithm

Time Complexity Space Complexity

Uniform Cost Search (UCS) $O(b^{(1+C^*/\epsilon)})$ $O(b^{(1+C^*/\epsilon)})$

Iterative Deepening Search (IDS) $O(b^d)$ $O(bd)$

Where:

- **b** = Branching factor
- **d** = Depth of solution
- **C*** = Cost of optimal solution
- **ϵ** = Minimum step cost

Relation to Intelligent Agents

1. Simple Reflex Agent

- Makes decisions based only on the current state.
- Has no memory of previously visited paths.
- May revisit the same locations repeatedly.

2. Model-Based Agent

- Maintains an internal map of explored paths.
- Remembers visited locations.
- Avoids repeated exploration.
- Performs better in unknown environments.

3. Goal-Based Agent

- Focuses on reaching the exit.
- Plans actions that move closer to the goal.
- Chooses paths based on achieving the objective.

4. Utility-Based Agent

- Evaluates multiple possible paths.
- Considers movement cost, safety, and efficiency.

- Selects the path with the highest overall benefit.

Effect of Agent Design on Decision-Making

The design of the intelligent agent significantly affects the robot's performance.

- **Simple Reflex Agents** react immediately but cannot learn from experience.
- **Model-Based Agents** remember explored areas and make better decisions.
- **Goal-Based Agents** plan their actions toward reaching the exit.
- **Utility-Based Agents** evaluate multiple factors, such as path cost and safety, before selecting the best action.

A more advanced agent design results in more efficient navigation and better decision-making.

Most Suitable Combination

The most effective solution is a **Model-Based Goal-Based Agent** combined with **Uniform Cost Search (UCS)**.

Justification

This combination is suitable because:

- The Model-Based Agent remembers explored paths and updates its internal map.
- The Goal-Based Agent continuously plans actions toward reaching the exit.
- Uniform Cost Search guarantees the minimum-cost path.
- The robot avoids unnecessary exploration and efficiently reaches the goal.

Conclusion

A robot navigating an unknown maze requires both an intelligent decision-making mechanism and an effective search strategy. While **IDS** is useful when memory is limited and the solution depth is unknown, **UCS** is preferred when finding the least-cost path is important. Combining a **Model-Based Goal-Based Agent** with **Uniform Cost Search** provides efficient exploration, optimal path selection, and reliable decision-making.

Result

Thus, **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)** are effective uninformed search strategies for solving the unknown maze problem. Among them, **Uniform Cost Search** combined with a **Model-Based Goal-Based Agent** is the most suitable approach because it remembers explored paths, minimizes total path cost, guarantees the optimal solution, and enables the robot to reach the exit efficiently.

